Table 1. Complete mathematical formulations showing CIA weights for each mission objective

Mission Objective	Title	Confidentiality Relative Weight	Integrity Relative Weight	Availability Relative Weight
Service Deployment	Establishing university services and systems, including network infrastructure, communications, etc.	0.3	0.5	0.5
Service Enhancement	Strengthening and expanding university services and systems.	0.35	0.3	0.3
Security Assurance	Ensuring the security of services and systems.	0.35	0.2	0.2

Table 2. Security Metric Formulas for Risk Assessment and Countermeasure Evaluation

Category	Formula	Description
New Total Risk	$NewTotalRisk = TotalRisk \times (1 - RiskReduction)$	Calculates the updated risk level after implementing a countermeasure
Reduced Risk	$ReducedRisk = TotalRisk - NewTotalRisk$	Calculates the amount of risk reduction achieved
Priority Score	$PriorityScore_i = W_{TPSC} \times TotalRisk_C + W_{TPSI} \times TotalRisk_I + W_{TPSA} \times TotalRisk_A$	Calculates priority score for each threat i using weighted CIA risks
Implementation Cost	$Cost(cm) = \alpha \times T(cm) + \beta \times \frac{R_{req}(cm)}{R_{avail}(D(cm))} \times I(D(cm))$	Cost calculation for countermeasure implementation
Negative Impact on QoS	$NI(cm) = DI(cm) + IDI(cm)$	Negative impact on quality of service
Direct Impact on QoS	$DI(cm) = (1 - P(cm)) \times \sum_{s \in S(cm)} I(s) \times T(cm) \times U(s)$	Direct impact on quality of service
Indirect Impact on QoS	$IDI(cm) = P(cm) \times \sum_{s \in S(cm)} I(s) \times T(cm) \times U(s)$	Indirect impact on quality of service
Threat Risk Reduction Coverage	$TRRC_C(cm) = \sum_{t \in T(cm)} RiskReduction_C(cm, t)$ $TRRC_I(cm) = \sum_{t \in T(cm)} RiskReduction_I(cm, t)$ $TRRC_A(cm) = \sum_{t \in T(cm)} RiskReduction_A(cm, t)$	Coverage of risk reduction for all threats by countermeasure
Total TRRC Score	$TRRC(cm) = W_{TRRC_C} \times TRRC_C(cm) + W_{TRRC_I} \times TRRC_I(cm) + W_{TRRC_A} \times TRRC_A(cm)$	Total threat risk reduction coverage score
Normalization	$Normalized\_Value = 1 + (N - 1) \times \frac{Value - Min\_Value}{Max\_Value - Min\_Value}$	Normalizes values to a scale of 1 to N

Normalized Cost Score	$CostScore = 1 + (N-1) \times \frac{Max\_Cost - Cost(cm)}{Max\_Cost - Min\_Cost}$	Normalizes cost scores (higher is better)
CM Utility Score	$CMUtilityScore(cm) = (W_{RR} \times RRScore(cm) + W_{TRRC} \times TRRCScore(cm)) - (W_{DC} \times CostScore(cm) + W_{QS} \times QSScore(cm))$	Overall utility score for countermeasure selection

Table 3. Security Model Variables and Their Definitions

Variable	Description
$W_{TPSC}, W_{TPSI}, W_{TPSA}$	Weights for Confidentiality, Integrity, and Availability aspects in threat priority calculation
$TotalRisk_C, TotalRisk_I, TotalRisk_A$	Total risk values for Confidentiality, Integrity, and Availability aspects
$T(cm)$	Implementation time for countermeasure cm
$R_{req}(cm)$	Resources required for countermeasure implementation
$R_{avail}\big(D(cm)\big)$	Available resources in device D where countermeasure will be deployed
$I\big(D(cm)\big)$	Importance of device D, where countermeasure will be deployed
$P(cm)$	Power of countermeasure to stop threat occurrence
$S(cm)$	Set of services affected by countermeasure cm
$I(s)$	Importance of service s (range 0-1)
$U(s)$	Number of users of service s affected by countermeasure
$W_{RR}, W_{DC}, W_{QS}, W_{TRRC}$	Weights for Risk Reduction, Deployment Cost, Quality of Service, and Threat Risk Reduction Coverage Criteria
N	Upper bound for normalization (set to 10 in the model)
$W_{TRRC_C}, W_{TRRC_I}, W_{TRRC_A}$	Weights for TRRC Aspects in Confidentiality, Integrity, and Availability

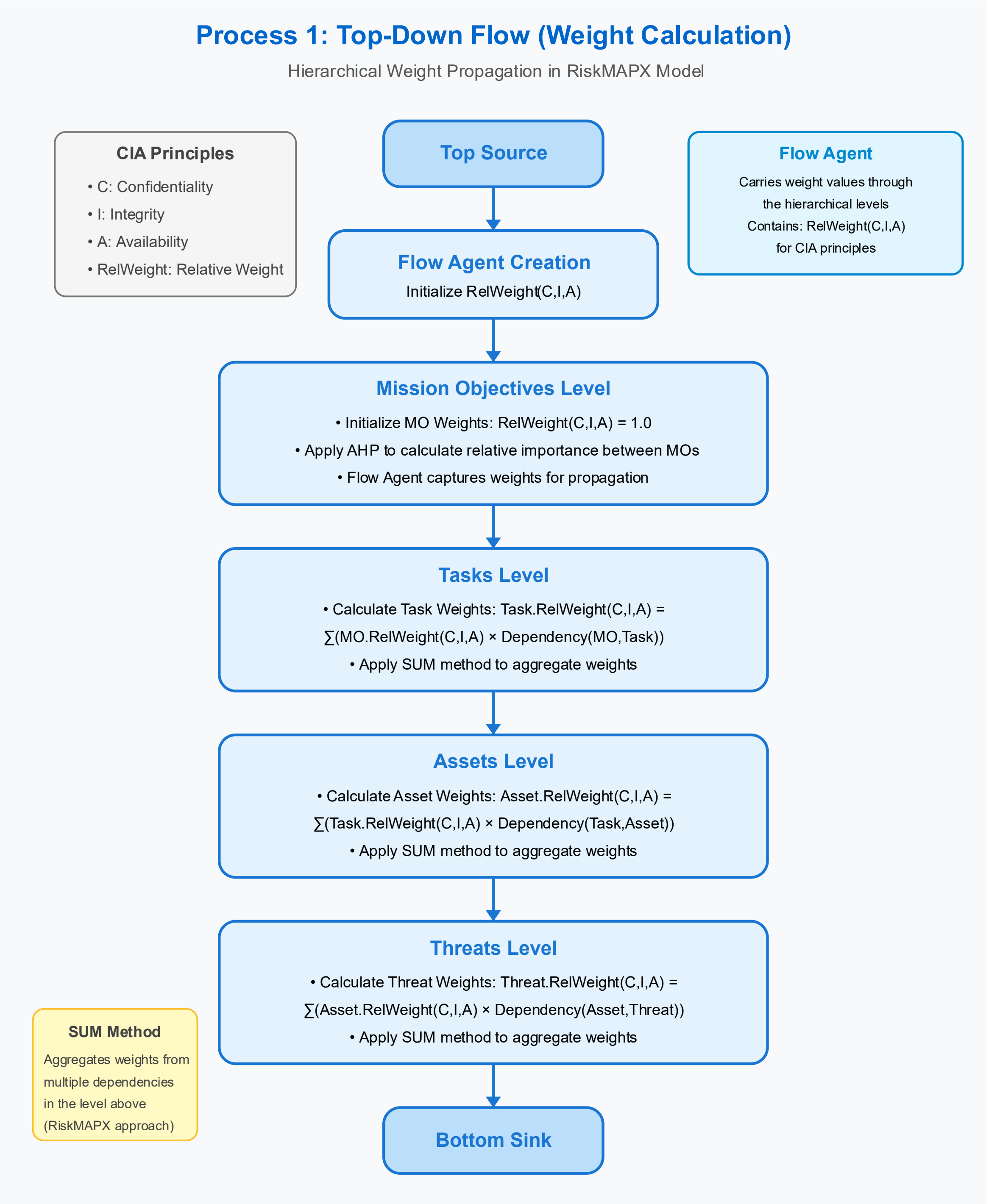

Figure 1. Top-Down Hierarchical Weight Propagation Workflow in the RiskMAPX Security Assessment Model

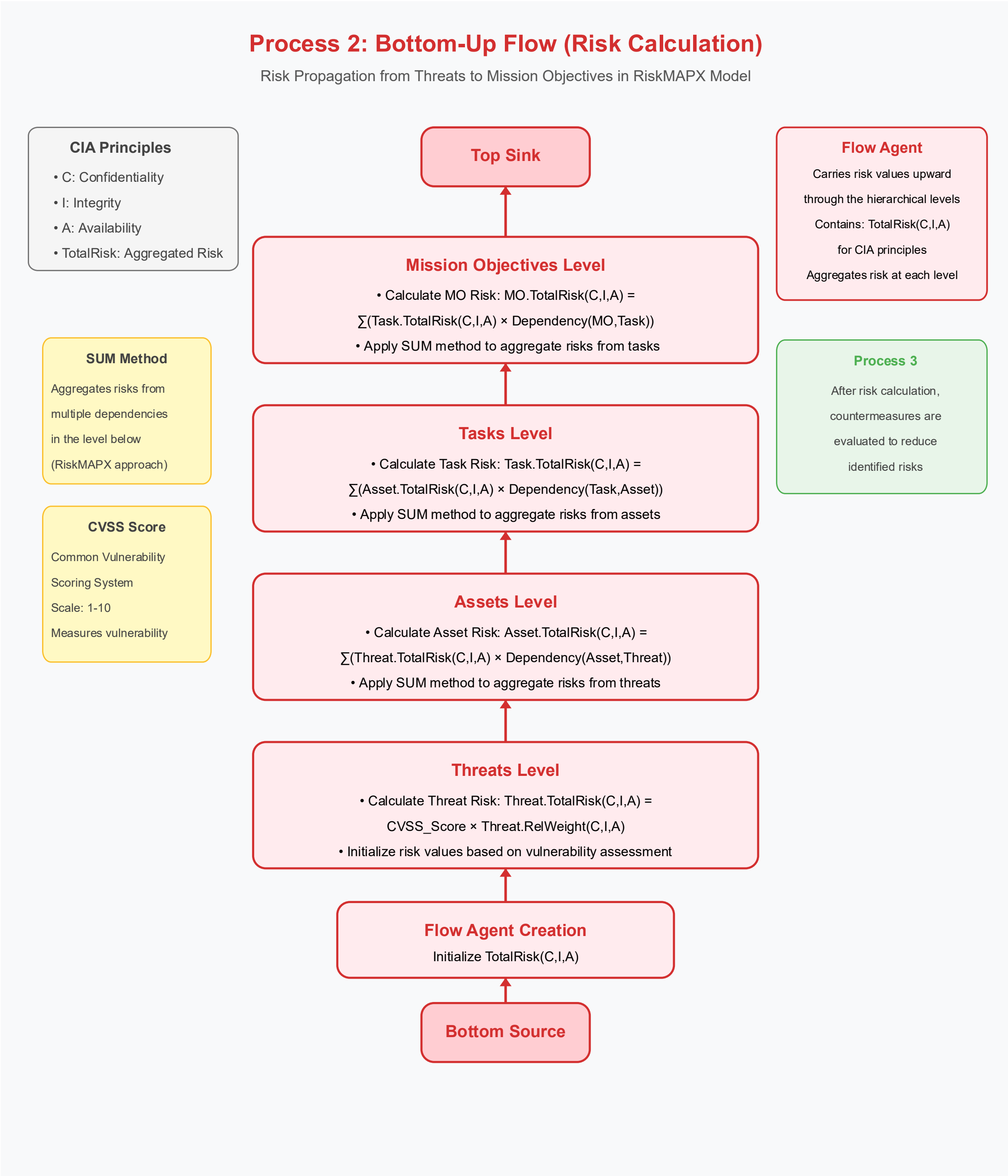

Figure 2. Bottom-Up Risk Aggregation Process in the RiskMAPX Security Assessment Model

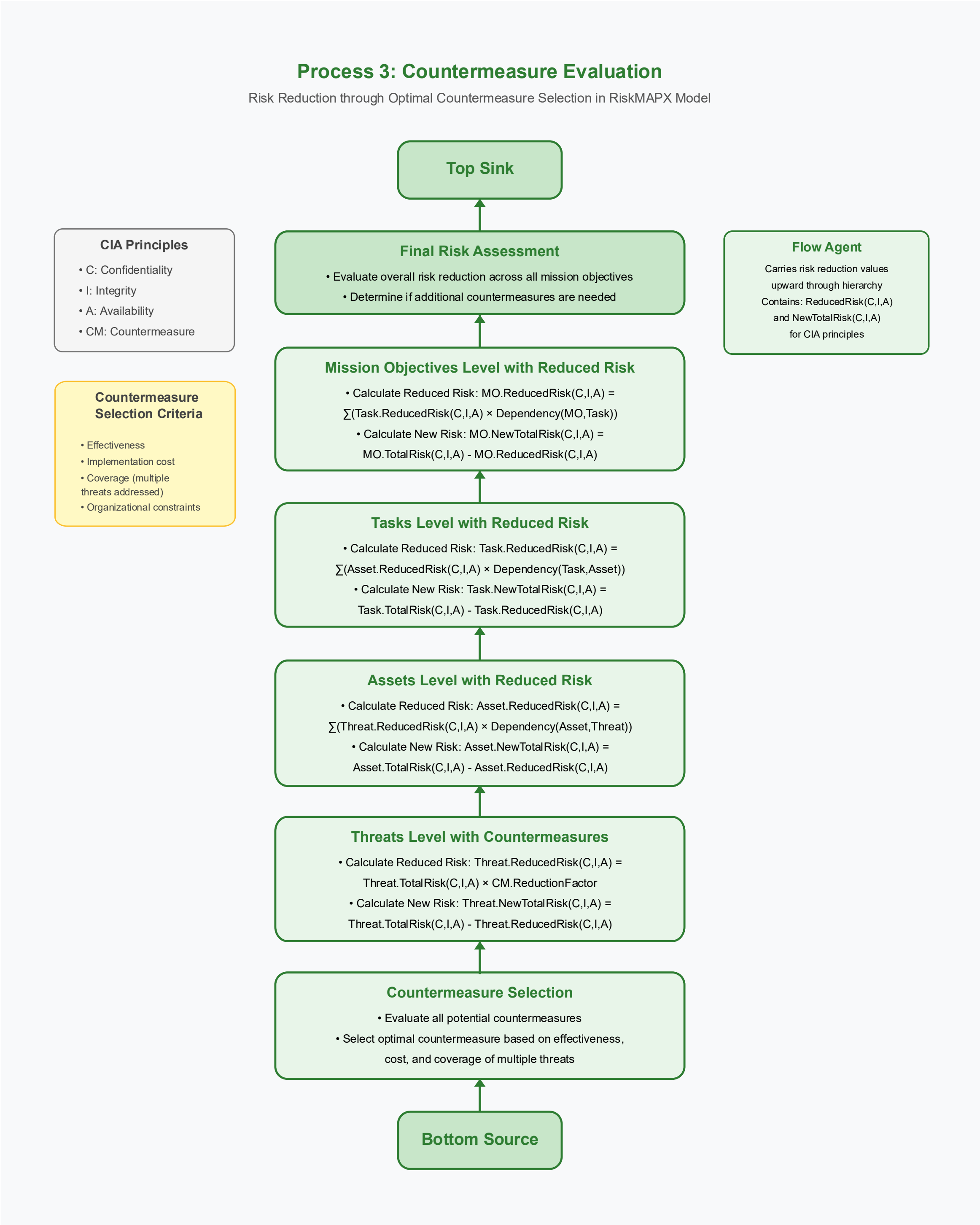

Figure 3. Countermeasure Evaluation and Risk Reduction Process in the RiskMAPX Framework

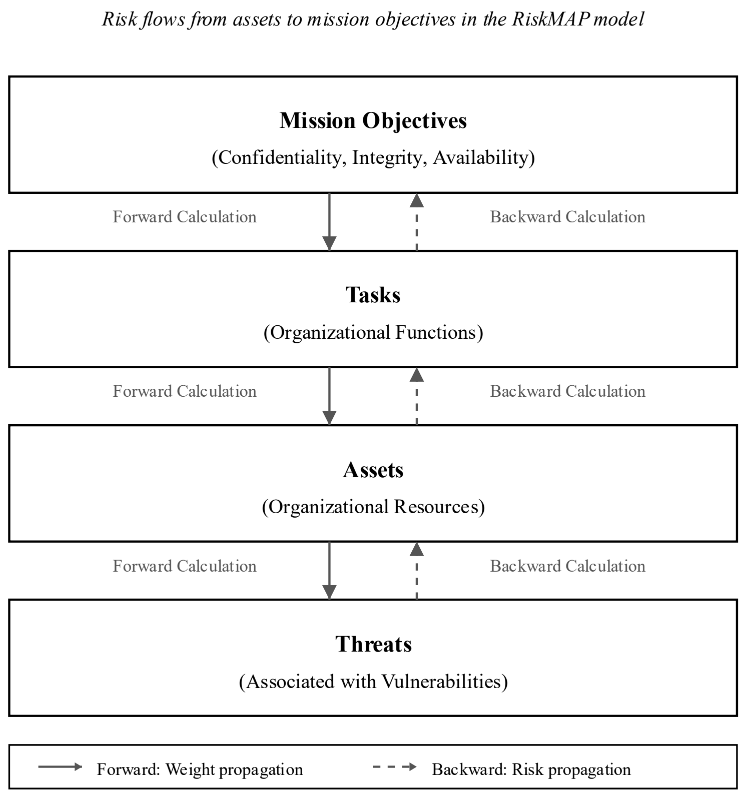

Figure 4. Bidirectional Risk and Weight Propagation in the RiskMAPX Framework

Table 4. Security Monitoring and Infrastructure Protection Task Definitions

ID	**Name**
Task1	Maintenance and development of virtual infrastructure
Task2	Backup of virtual infrastructure
Task3	Monitoring of virtual infrastructure
Task4	Review logs and events of virtual infrastructure.
Task5	Implementation of cooling standards for data centers
Task6	Implementation of cabling standards for data centers
Task7	Implementation of backup power standards
Task8	Implementation of fire suppression standards
Task9	Creation and examination of network topology
Task10	Creation of VLANs
Task11	Securing communications between VLANs
Task12	Monitoring and reviewing logs of switches and routers
Task13	Creating backup files of configurations
Task14	Establishing BGP communication

ID	Name
Task15	Maintenance and configuration of edge firewalls
Task16	Maintenance and configuration of edge routers
Task17	Management and maintenance of the main Web Server2
Task18	Management and maintenance of hosting servers
Task19	Management and maintenance of DNS service
Task20	Backup creation
Task21	Reviewing and monitoring logs and events
Task22	Management and maintenance of security equipment
Task23	Security monitoring and log analysis using Splunk
Task24	Monitoring system resources using Zabbix
Task25	Periodic penetration testing
Task26	Vulnerability scanning with Nessus
Task27	Hardening and securing network switches
Task28	Securing DNS services
Task29	Securing web servers
Task30	Updating frameworks and applying security patches

Table 5. Critical Network and Infrastructure Assets Identified for Protection Assessment

ID	Name
Asset1	Edge router
Asset2	External router
Asset3	External firewall
Asset4	Internal firewall
Asset5	External DNS
Asset6	Internal DNS
Asset7	vCenter
Asset8	vSwitch
Asset9	Machine 1
Asset10	Machine 2
Asset11	Web Server 1
Asset12	Web Server 2
Asset13	Web Application Firewall

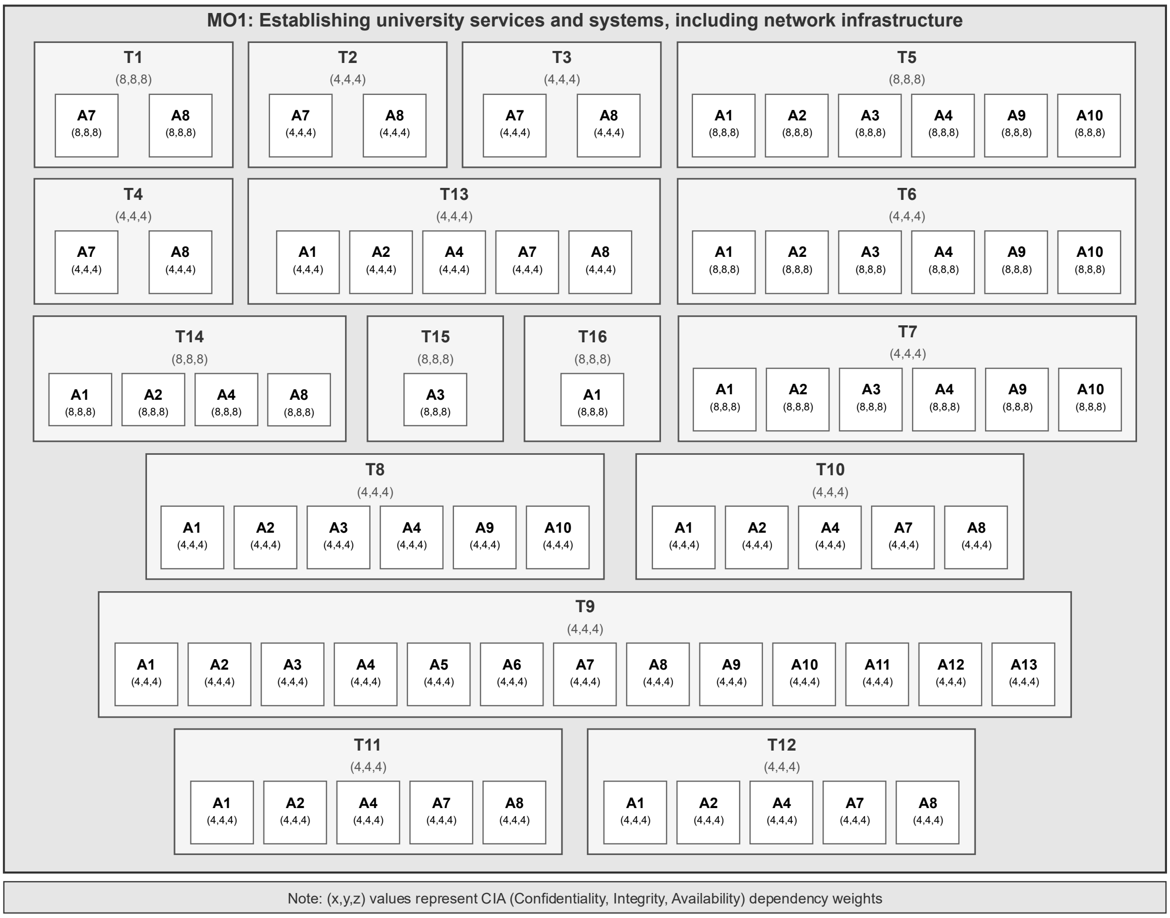

Figure 5. Hierarchical Dependency Representation of Mission Objective Service Deployment for Case Study

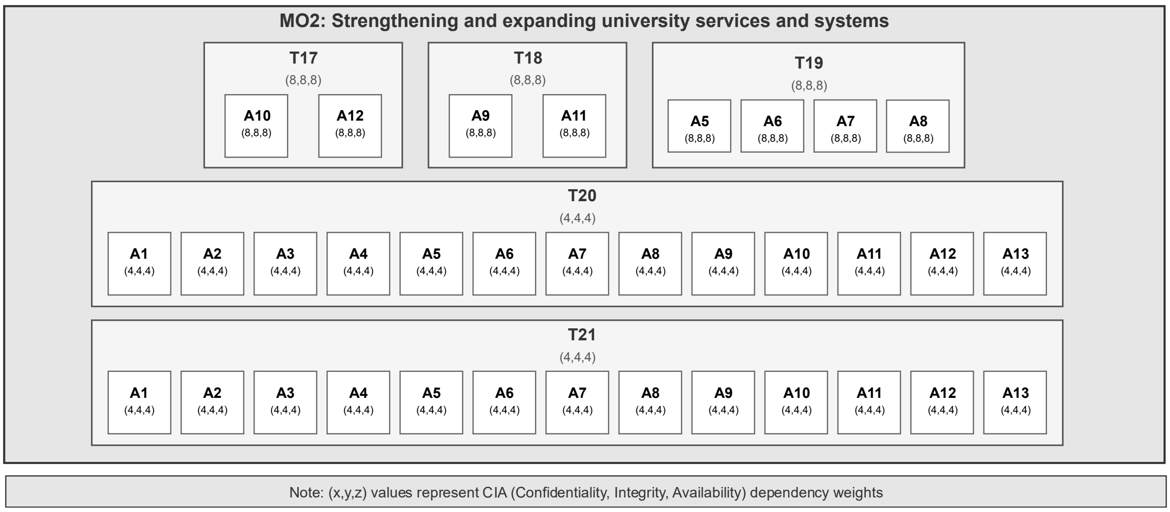

Figure 6. Hierarchical Dependency Representation of Mission Objective Service Enhancement for Case Study

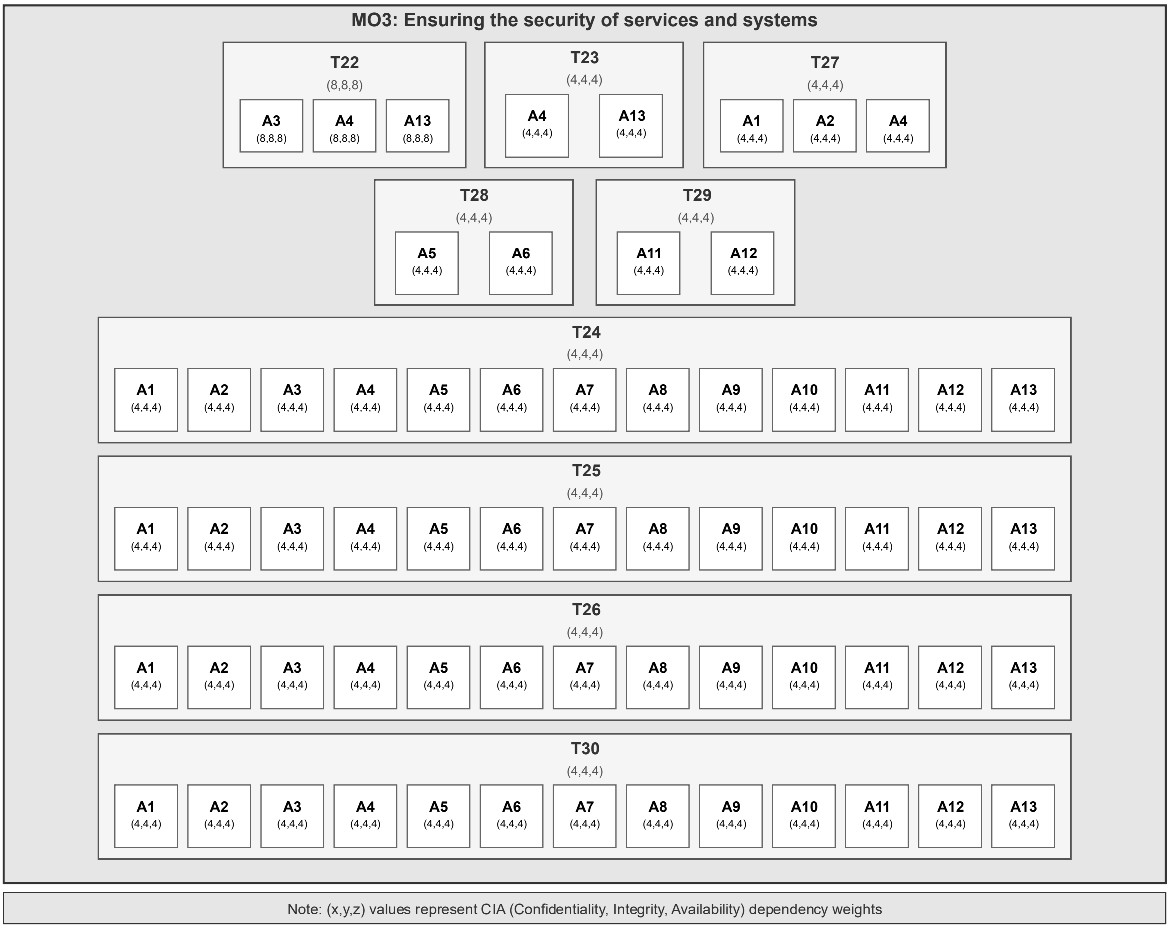

Figure 7. Hierarchical Dependency Representation of Mission Objective Security Assurance for Case Study

Table 6. Vulnerability Assessment Matrix with Associated CVSS Scores and Impact Ratings

ID	CVE Id	Name	CVSS Score C	CVSS Score I	CVSS Score A	Impact Score C	Impact Score I	Impact Score A	Likelihood Score
Vul1	CVE-2021-40438	Apache 2.4.x < 2.4.54 Multiple Vulnerabilities	7.5	7.5	7.5	10	10	10	5
Vul2	CVE-2022-1292	OpenSSL 1.1.1 < 1.1.1o Vulnerability	10	10	10	10	10	10	5
Vul3	CVE-2015-4000	SSL/TLS Diffie-Hellman Modulus <= 1024	4.3	4.3	4.3	6	6	6	2

ID	CVE Id	Name	CVSS Score C	CVSS Score I	CVSS Score A	Impact Score C	Impact Score I	Impact Score A	Likelihood Score
		Bits (Logjam)							
Vul4	CVE-2004-2761	SSL Certificate Signed Using Weak Hashing Algorithm	5	5	5	10	3	3	2
Vul5	CVE-2013-2566	SSL RC4 Cipher Suites Supported (Bar Mitzvah)	4.3	4.3	4.3	3	3	3	2
Vul6	CVE-2016-2183	SSL Medium Strength Cipher Suites Supported (SWEET32)	5	5	5	6	6	6	2

Table 7. Threat Characterization and Vulnerability Association Matrix

ID	Name	Description	Related Vulnerability
Threat1	Server-Side Request Forgery (SSRF)	Threat actors can exploit the CVE-2021-40438 vulnerability to trigger SSRF attacks by sending specially crafted requests to the Apache HTTP Server with the mod proxy module enabled	CVE-2021-40438
Threat2	Remote Code Execution	An attacker could exploit the vulnerability to execute arbitrary commands on the system.	CVE-2022-1292
Threat3	Unauthorized Access	An attacker could gain unauthorized access to sensitive information on the affected system.	CVE-2022-1292
Threat4	Downgrade of DH parameter curves to export strength	The core attack is a downgrade of the DH parameters used during the initial TLS setup to an export strength of 512 bits. This allows attackers to break the DH key exchange with a 512-bit parameter set in near real-time	CVE-2015-4000
Threat5	Spoofing Attacks	The vulnerability allows context-dependent attackers to conduct spoofing attacks by exploiting the MD5 algorithm in the signature algorithm of an X.509 certificate.	CVE-2004-2761
Threat6	Plaintext-recovery attacks	Remote attackers can potentially recover specific plaintext data from an arbitrary ciphertext block in an SSH session.	CVE-2013-2566
Threat7	Birthday Attack	The vulnerability allows remote attackers to attack a birthday against a long-duration encrypted session, potentially obtaining cleartext data.	CVE-2016-2183

Table 8. Countermeasure Analysis with Implementation Status, Cost, and Operational Impact

ID	Name	Has Been Deployed	Total Cost	Negative Impact on QoS
Countermeasure1	Upgrade to the Latest Version of Apache	FALSE	500	2
Countermeasure2	Disable or Limit mod proxy Module	FALSE	200	2
Countermeasure3	Implement Web Application Firewalls (WAF)	TRUE	2000	3
Countermeasure4	Input Validation and Filtering	FALSE	100	2
Countermeasure5	Implement Network Segmentation	FALSE	1	1
Countermeasure6	Apply the latest fixes of OpenSSL	FALSE	3000	10
Countermeasure7	Upgrade to supported versions of OpenSSL	FALSE	3000	10
Countermeasure8	Implement strong access controls	FALSE	3000	10
Countermeasure9	Monitor network traffic	FALSE	3000	10
Countermeasure10	Educate users	FALSE	3000	10
Countermeasure11	Upgrade SSL client applications and libraries	FALSE	1000	3
Countermeasure12	Use of specific and static DH parameter sets	FALSE	100	2
Countermeasure13	Upgrade to a Secure Hash Algorithm	FALSE	500	3
Countermeasure14	Replace MD5 Certificates	FALSE	2000	3
Countermeasure15	Implement Certificate Authorities (CAs) Best Practices	FALSE	2000	3
Countermeasure16	Upgrade to a secure version of TLS and SSL	FALSE	500	2
Countermeasure17	Disable RC4 cipher suites	FALSE	200	2
Countermeasure18	Use strong encryption algorithms such as AES	FALSE	1000	3
Countermeasure19	Disable 3DES Cipher Suites	FALSE	200	2

Table 9. Risk Reduction Effectiveness of Countermeasures Against Identified Threats Across CIA Triad Dimensions

CM ID	Threats ID	Risk Reduction C	Risk Reduction I	Risk Reduction A
Countermeasure1	Threat1	0.8	0.7	0.9
Countermeasure2	Threat1	0.6	0.5	0.8
Countermeasure3	Threat1	0.9	0.8	0.7
Countermeasure4	Threat1	0.7	0.6	0.5
Countermeasure5	Threat1	0.8	0.9	0.7
Countermeasure5	Threat2	0.7	0.8	0.9
Countermeasure6	Threat2	0.6	0.7	0.8
Countermeasure7	Threat2	0.9	0.6	0.7
Countermeasure8	Threat2	0.8	0.7	0.6
Countermeasure9	Threat2	0.7	0.9	0.8
Countermeasure10	Threat2	0.9	0.8	0.7
Countermeasure5	Threat3	0.9	0.9	0.9
Countermeasure6	Threat3	0.1	0.1	0.1

Countermeasure7	Threat3	0.1	0.1	0.1
Countermeasure8	Threat3	0.1	0.1	0.1
Countermeasure9	Threat3	0.1	0.1	0.1
Countermeasure10	Threat3	0.1	0.1	0.1
Countermeasure11	Threat4	0.9	0.6	0.7
Countermeasure12	Threat4	0.8	0.7	0.6
Countermeasure13	Threat5	0.7	0.9	0.8
Countermeasure14	Threat5	0.6	0.8	0.9
Countermeasure15	Threat5	0.8	0.7	0.6
Countermeasure16	Threat6	0.7	0.6	0.8
Countermeasure17	Threat6	0.9	0.8	0.7
Countermeasure18	Threat6	0.8	0.7	0.9
Countermeasure19	Threat7	0.6	0.9	0.8

Table 10. Mission-Critical Task Importance Ratings Across Confidentiality, Integrity, and Availability Security Attributes

Block Id	Importance C	Importance I	Importance A
MO1_T1	8	8	8
MO1_T2	4	4	4
MO1_T3	4	4	4
MO1_T4	4	4	4
MO1_T5	8	8	8
MO1_T6	8	8	8
MO1_T7	8	8	8
MO1_T8	4	4	4
MO1_T9	4	4	4
MO1_T10	4	4	4
MO1_T11	4	4	4
MO1_T12	4	4	4
MO1_T13	4	4	4
MO1_T14	8	8	8
MO1_T15	8	8	8
MO1_T16	8	8	8
MO2_T17	8	8	8
MO2_T18	8	8	8
MO2_T19	8	8	8
MO2_T20	4	4	4
MO2_T21	4	4	4
MO3_T22	8	8	8

Block Id	Importance C	Importance I	Importance A
MO3_T23	4	4	4
MO3_T24	4	4	4
MO3_T25	4	4	4
MO3_T26	4	4	4
MO3_T27	4	4	4
MO3_T28	4	4	4
MO3_T29	4	4	4
MO3_T30	4	4	4
T1_A7	8	8	8
T1_A8	8	8	8
T2_A7	4	4	4
T2_A8	4	4	4
T3_A7	4	4	4
T3_A8	4	4	4
T4_A7	4	4	4
T4_A8	4	4	4
T5_A1	8	8	8
T5_A2	8	8	8
T5_A3	8	8	8
T5_A4	8	8	8
T5_A9	8	8	8
T5_A10	8	8	8
T6_A1	8	8	8
T6_A2	8	8	8
T6_A3	8	8	8
T6_A4	8	8	8
T6_A9	8	8	8
T6_A10	8	8	8
T7_A1	8	8	8
T7_A2	8	8	8
T7_A3	8	8	8
T7_A4	8	8	8
T7_A9	8	8	8
T7_A10	8	8	8
T8_A1	4	4	4
T8_A2	4	4	4
T8_A3	4	4	4
T8_A4	4	4	4

Block Id	Importance C	Importance I	Importance A
T8_A9	4	4	4
T8_A10	4	4	4
T9_A1	4	4	4
T9_A2	4	4	4
T9_A3	4	4	4
T9_A4	4	4	4
T9_A5	4	4	4
T9_A6	4	4	4
T9_A7	4	4	4
T9_A8	4	4	4
T9_A9	4	4	4
T9_A10	4	4	4
T9_A11	4	4	4
T9_A12	4	4	4
T9_A13	4	4	4
T10_A1	4	4	4
T10_A2	4	4	4
T10_A4	4	4	4
T10_A7	4	4	4
T10_A8	4	4	4
T11_A1	4	4	4
T11_A2	4	4	4
T11_A4	4	4	4
T11_A7	4	4	4
T11_A8	4	4	4
T12_A1	4	4	4
T12_A2	4	4	4
T12_A4	4	4	4
T12_A7	4	4	4
T12_A8	4	4	4
T13_A1	4	4	4
T13_A2	4	4	4
T13_A4	4	4	4
T13_A7	4	4	4
T13_A8	4	4	4
T14_A1	8	8	8
T14_A2	8	8	8
T14_A4	8	8	8

Block Id	Importance C	Importance I	Importance A
T14_A8	8	8	8
T15_A3	8	8	8
T16_A1	8	8	8
T17_A10	8	8	8
T17_A12	8	8	8
T18_A9	8	8	8
T18_A11	8	8	8
T19_A5	8	8	8
T19_A6	8	8	8
T19_A7	8	8	8
T19_A8	8	8	8
T20_A1	4	4	4
T20_A2	4	4	4
T20_A3	4	4	4
T20_A4	4	4	4
T20_A5	4	4	4
T20_A6	4	4	4
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T21_A1	4	4	4
T21_A2	4	4	4
T21_A3	4	4	4
T21_A4	4	4	4
T21_A5	4	4	4
T21_A6	4	4	4
T21_A7	4	4	4
T21_A8	4	4	4
T21_A9	4	4	4
T21_A10	4	4	4
T21_A11	4	4	4
T21_A12	4	4	4
T21_A13	4	4	4
T22_A3	8	8	8

Block Id	Importance C	Importance I	Importance A
T22_A4	8	8	8
T22_A13	8	8	8
T23_A4	4	4	4
T23_A13	4	4	4
T24_A1	4	4	4
T24_A2	4	4	4
T24_A3	4	4	4
T24_A4	4	4	4
T24_A5	4	4	4
T24_A6	4	4	4
T24_A7	4	4	4
T24_A8	4	4	4
T24_A9	4	4	4
T24_A10	4	4	4
T24_A11	4	4	4
T24_A12	4	4	4
T24_A13	4	4	4
T25_A1	4	4	4
T25_A2	4	4	4
T25_A3	4	4	4
T25_A4	4	4	4
T25_A5	4	4	4
T25_A6	4	4	4
T25_A7	4	4	4
T25_A8	4	4	4
T25_A9	4	4	4
T25_A10	4	4	4
T25_A11	4	4	4
T25_A12	4	4	4
T25_A13	4	4	4
T26_A1	4	4	4
T26_A2	4	4	4
T26_A3	4	4	4
T26_A4	4	4	4
T26_A5	4	4	4
T26_A6	4	4	4
T26_A7	4	4	4
T26_A8	4	4	4

Block Id	Importance C	Importance I	Importance A
T26_A9	4	4	4
T26_A10	4	4	4
T26_A11	4	4	4
T26_A12	4	4	4
T26_A13	4	4	4
T27_A1	4	4	4
T27_A2	4	4	4
T27_A4	4	4	4
T28_A5	4	4	4
T28_A6	4	4	4
T29_A11	4	4	4
T29_A12	4	4	4
T30_A1	4	4	4
T30_A2	4	4	4
T30_A3	4	4	4
T30_A4	4	4	4
T30_A5	4	4	4
T30_A6	4	4	4
T30_A7	4	4	4
T30_A8	4	4	4
T30_A9	4	4	4
T30_A10	4	4	4
T30_A11	4	4	4
T30_A12	4	4	4
T30_A13	4	4	4
A1_Th4	8	8	8
A1_Th5	6	6	6
A1_Th6	5	5	5
A1_Th7	7	7	7
A2_Th4	7	7	7
A2_Th5	6	6	6
A2_Th6	4	4	4
A2_Th7	5	5	5
A3_Th4	9	9	9
A3_Th5	8	8	8
A3_Th6	7	7	7
A3_Th7	6	6	6
A4_Th4	9	9	9

Block Id	Importance C	Importance I	Importance A
A4_Th5	7	7	7
A4_Th6	6	6	6
A4_Th7	8	8	8
A5_Th1	7	7	7
A5_Th2	9	9	9
A6_Th1	6	6	6
A6_Th2	8	8	8
A7_Th2	9	9	9
A7_Th4	9	9	9
A7_Th5	8	8	8
A7_Th6	7	7	7
A7_Th7	7	7	7
A8_Th1	6	6	6
A8_Th2	8	8	8
A9_Th1	7	7	7
A9_Th2	9	9	9
A9_Th3	6	6	6
A9_Th4	8	8	8
A9_Th5	7	7	7
A9_Th6	6	6	6
A9_Th7	7	7	7
A10_Th1	5	5	5
A10_Th2	8	8	8
A10_Th3	6	6	6
A10_Th4	7	7	7
A10_Th5	6	6	6
A10_Th6	5	5	5
A10_Th7	6	6	6
A11_Th1	6	6	6
A11_Th2	9	9	9
A11_Th3	7	7	7
A11_Th4	8	8	8
A11_Th5	7	7	7
A11_Th6	6	6	6
A11_Th7	6	6	6
A12_Th1	5	5	5
A12_Th2	8	8	8
A12_Th3	6	6	6

Block Id	Importance C	Importance I	Importance A
A12_Th4	7	7	7
A12_Th5	6	6	6
A12_Th6	5	5	5
A12_Th7	6	6	6
A13_Th4	7	7	7
A13_Th5	8	8	8
A13_Th6	6	6	6
A13_Th7	7	7	7

Table 11. Decision Parameters for Security Implementation

Title of Decision Parameter	**Parameter ID**	**Value**
Maximum Budget	MaxBudget	5,000$
Weight for Risk Reduction in Confidentiality	W_RR_C	0.1
Weight for Risk Reduction in Integrity	W_RR_I	0.4
Weight for Risk Reduction in Availability	W_RR_A	0.5
Weight for Risk Reduction	W_RR	0.4
Weight for Deployment Cost	W_DC	0.2
Weight for Quality of Service	W_QS	0.3
Weight for Total Risk Reduction in Confidentiality	W_TRRC	0.1
Weight for Total Risk Reduction Cost in Confidentiality	W_TRRCC	0.4
Weight for Total Risk Reduction Cost in Integrity	W_TRRCI	0.3
Weight for Total Risk Reduction Cost in Availability	W_TRRCA	0.3
Weight for Total Protection Score in Confidentiality	W_TPSC	0.1
Weight for Total Protection Score in Integrity	W_TPSI	0.4
Weight for Total Protection Score in Availability	W_TPSA	0.5

Table 12. Risk Reduction Percentages by Mission Objective

Mission Objective	**Risk Reduction in Availability (%)**	**Risk Reduction in Integrity (%)**	**Risk Reduction in Confidentiality (%)**
Service Deployment	97%	97%	97%
Service Enhancement	85%	85%	82%
Security Assurance	89%	89%	87%

Table 13. Countermeasure Effectiveness and Risk Impact Analysis

Countermeasure	**TotalRiskC**	**TotalRiskI**	**TotalRiskA**	**NewTotalRiskC**	**NewTotalRiskI**	**NewTotalRiskA**	**Cost ($)**

Cm5	Threat3	10	10	10	3	4	4	3000
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	2.203×10^{7}	1.431×10^{7}	1.431×10^{7}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	4.292×10^{7}	2.830×10^{7}	2.830×10^{7}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	2.776×10^{8}	1.831×10^{8}	1.831×10^{8}	
Cm6	Threat3	10	10	10	2	1	1	100
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	2.459×10^{7}	1.84×10^{7}	1.84×10^{7}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	4.821×10^{7}	3.598×10^{7}	3.598×10^{7}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	3.14×10^{8}	2.344×10^{8}	2.344×10^{8}	
Cm7	Threat3	10	10	10	4	3	3	500
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	6.494×10^{6}	2.469×10^{7}	2.469×10^{7}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	1.332×10^{7}	4.857×10^{7}	4.857×10^{7}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	8.625×10^{7}	3.162×10^{8}	3.162×10^{8}	
Cm8	Threat3	10	10	10	1	2	2	1000
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	1.23×10^{7}	1.849×10^{7}	8.063×10^{8}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	2.411×10^{7}	3.634×10^{7}	2.673×10^{8}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	1.57×10^{8}	2.366×10^{8}	2.401×10^{9}	
Cm9	Threat3	10	10	10	3	4	4	1000
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	1.859×10^{7}	6.494×10^{6}	6.494×10^{6}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	3.67×10^{7}	1.332×10^{7}	1.332×10^{7}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	2.388×10^{8}	8.625×10^{7}	8.625×10^{7}	
Cm10	Threat3	10	10	10	2	3	3	500
	Service Deployment	8.063×10^{8}	6.783×10^{8}	6.783×10^{8}	6.297×10^{6}	1.249×10^{7}	1.249×10^{7}	
	Service Enhancement	2.673×10^{8}	2.468×10^{8}	2.468×10^{8}	1.26×10^{7}	2.483×10^{7}	2.483×10^{7}	
	Security Assurance	2.401×10^{9}	2.134×10^{9}	2.134×10^{9}	8.182×10^{7}	1.614×10^{8}	1.614×10^{8}	

Table 14. Countermeasure Cost-Benefit Analysis

Countermeasure	**Cost ($)**	**Risk Reduction**			**Avg Risk Reduction**	**Benefit/Cost Ratio**
		Confidentiality	Integrity	Availability		
Cm5	3,000	89%	90%	90%	89.7%	0.030
Cm6	100	87%	87%	87%	87.0%	0.870
Cm7	500	86%	85%	85%	85.3%	0.171

Countermeasure	Cost ($)		Risk Reduction		Avg Risk Reduction	Benefit/Cost Ratio
Cm8	1,000	88%	87%	87%	87.3%	0.087
Cm9	1,000	88%	89%	89%	88.7%	0.089
Cm10	500	87%	87%	87%	87.0%	0.174